Coherent nonlinear Thomson scattering of Laguerre–Gauss laser beams

Progress report

31 October 2026

We calculate the radiation emitted when an intense, circularly polarized Laguerre–Gauss (LG) laser pulse collides head-on with a relativistic electron beam. The radiation of all electrons is added coherently, as a complex field on a detector screen far downstream. From this field we compute the energy and angular momentum (spin and orbital) carried by each harmonic. We then ask how close each harmonic is to a single LG mode, and with which waist.

The report has two parts, the first two steps of the plan for the electron beam:

- **Part I (Sec. 1–5): a flat electron disk**, the ideal, fully coherent case. The results come from four simulations: a Gaussian laser (LG_0^0) and an LG_2^2 laser, each at two field strengths. A second set of tests, with electrons at rest, explains the lens-like effect the emitted beams show (Sec. 4.4). Sec. 5 repeats the four simulations with the electrons on a regular lattice instead of at random, which removes the sampling noise.
- **Part II (Sec. 6): a Gaussian bunch length**, which shows how coherence is lost. The disk is given a thickness, and we check how the field of Part I changes.

The third step, a bunch with ponderomotive microbunching that regains some of the coherence, is still to come.

Part I: a flat electron disk

1. Setup

Laser	$\lambda_L = 10$ μm, circular polarization, propagating along $-z$; LG_p^l mode with waist $w_0 = 25\,\lambda_L$ at the interaction point; Gaussian temporal envelope, 200 cycles in total (6.7 ps)
Field strength	$a_0 = 0.1$ and 0.2
Electrons	65 536 electrons at random positions in a thin disk perpendicular to z (radius $46.5\,\lambda_L$ for LG_0^0, $80.5\,\lambda_L$ for LG_2^2, just larger than the mode), all with $p_z = 13.48\,mc$ ($\gamma \approx 13.5$), moving along $+z$
Detector	flat circular screen facing the beam, at $Z = 4 \times 10^6\,\lambda_L = 40$ m; 128 radial $\times$ 64 azimuthal points; angular radius 94 μrad ($a_0 = 0.1$) or 156 μrad ($a_0 = 0.2$)
Harmonics	$N = 1$ and 2, at $\omega_1 \approx 715$–$726\,\omega_L$ ($\lambda_1 \approx 14$ nm, XUV)

The four runs:

Run	Laser mode	a_0	Screen radius
A	LG_0^0 (Gaussian)	0.1	$15\,w_0$
B	LG_0^0 (Gaussian)	0.2	$25\,w_0$
C	LG_2^2	0.1	$15\,w_0$
D	LG_2^2	0.2	$25\,w_0$

Sampling the electron beam. A real bunch has 10^8–10^{10} electrons, and we can follow 10^3–10^5 trajectories. The coherent field is a sum over the electrons, in effect an integral of the single-electron field over the disk. When the simulated electrons are placed at random, as in runs A–D, the sum also contains an artificial incoherent part. The random positions leave a speckle background whose relative power is of order (number of modes the screen resolves)/(number of simulated electrons), spread over all l. Measured against the lattice runs of Sec. 5, it is 0.2–1.4% of the power at $N = 1$ and 3–10% at $N = 2$ in runs A–D. A real bunch has 10^3–10^5 times more electrons, so its own background is that much weaker. The speckle is therefore a sampling artefact, not physics.

For the tests of Sec. 4.4 and the repeat of runs A–D in Sec. 5 we place the electrons on a regular lattice instead. There is one electron at the centre, and concentric rings with the same spacing d in radius and along each ring, so every electron stands for the same area. This turns the sum into a quadrature of the integral rather than a random sample, so it has no speckle. It is accurate when two conditions hold:

- d resolves the laser profile; we use $d \approx w_0/10$;
- the lattice's diffraction orders, at angles of about λ_N/d, fall outside the screen.

With $d = 2.4\,\lambda_L$ the disk of runs A and B (radius $1.86\,w_0$) needs 1 076 electrons instead of 65 536, about 60 times fewer. Run time scales with the electron count: the tests of Sec. 4.4 take 80 s each on 256 cores. Halving d changes the field by about 1%. In a separate check, the lattice field agreed with a 65 536-electron random run to within that run's speckle (0.1% of the power at $N = 1$, 0.7% at $N = 2$).

The interaction is classical and contains no recoil. Each electron's trajectory follows from the Lorentz force in the laser field. The radiation is the Liénard–Wiechert field of that trajectory.

2. The emitted field

2.1 Faraday tensor in the frequency domain

The basic output is the Fourier transform of the electromagnetic field tensor $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$ at every screen point $\mathbf{x}$ and at a chosen frequency ω:

$$F^{\alpha\beta}(\omega, \mathbf{x}) = \frac{1}{2\pi} \int dt\; e^{i\omega t}\, F^{\alpha\beta}(t, \mathbf{x}), \qquad E_i = c\, F^{i0}, \quad \mathbf{B} = (F^{32},\, F^{13},\, F^{21}).$$

For one electron of charge q, the time integral can be rewritten as an integral over the electron's proper time τ. We use the retarded-time relation $ct = r_0^0(\tau) + |\mathbf{x} - \mathbf{r}_0(\tau)|$:

$$F^{\alpha\beta}(\omega, \mathbf{x}) = \frac{q}{8\pi^2 \epsilon_0 c^2} \int d\tau\; e^{ik\,(r_0^0(\tau) + |\mathbf{R}(\tau)|)}\, K^{\alpha\beta}(\tau),$$

$$K^{\alpha\beta} = \frac{(u\cdot n)(n^\alpha w^\beta - n^\beta w^\alpha) - (w\cdot n)(n^\alpha u^\beta - n^\beta u^\alpha)}{|\mathbf{R}|\,(u\cdot n)^2} + \frac{c^2\,(n^\alpha u^\beta - n^\beta u^\alpha)}{|\mathbf{R}|^2\,(u\cdot n)^2},$$

where $k = \omega/c$, $\mathbf{R}(\tau) = \mathbf{x} - \mathbf{r}_0(\tau)$ is the vector from the electron to the observation point, $n = (1, \mathbf{R}/|\mathbf{R}|)$ is the null vector along it, $u^\mu = dr_0^\mu/d\tau$ is the four-velocity and $w^\mu = du^\mu/d\tau$ the four-acceleration. The first term is the radiation field ($\propto 1/R$). The second is the velocity field ($\propto 1/R^2$), which is negligible on a screen 40 m away.

The total field is the **coherent sum over all electrons**, $F = \sum_j F_j$. Every electron's contribution carries its own phase $e^{ik(r_0^0 + |\mathbf{R}|)}$. Interference between electrons therefore decides the transverse structure of the emitted beam. It is not just the sum of single-electron patterns.

2.2 Frequencies

We evaluate the field at the harmonics of nonlinear Thomson scattering in the direction of the screen normal ($+z$):

$$\omega_N = N\,\omega_L\,\frac{q^0 - \mathbf{n}_L\cdot\mathbf{q}}{q^0 - \mathbf{n}\cdot\mathbf{q}}, \qquad q^\mu = p^\mu + \frac{m^2c^2\langle a^2\rangle}{2\,(k_L\cdot p)}\,k_L^\mu, \quad \langle a^2\rangle = \frac{a_0^2}{2},$$

with q the electron's quasi-momentum in the wave, $\mathbf{n}_L$ the laser direction and $\mathbf{n}$ the observation direction. A larger a_0 gives a larger effective mass and so a slightly lower frequency: $\omega_1/\omega_L = 725.5$ at $a_0 = 0.1$ and 714.9 at $a_0 = 0.2$. Each harmonic is evaluated at this one, on-axis frequency over the whole screen.

2.3 What the field looks like

The exported field contains the spherical carrier e^{ikR} of the outgoing wave. At $\lambda_1 \approx 14$ nm and $R = 40$ m this gives tens of fringes across the screen, which hide the structure of the beam (Fig. 1, left). For display we multiply by $e^{-ik(R-Z)}$, which removes the carrier and keeps everything else (Fig. 1, right). This changes only how the field is plotted, not the physics: every observable below uses the full field.

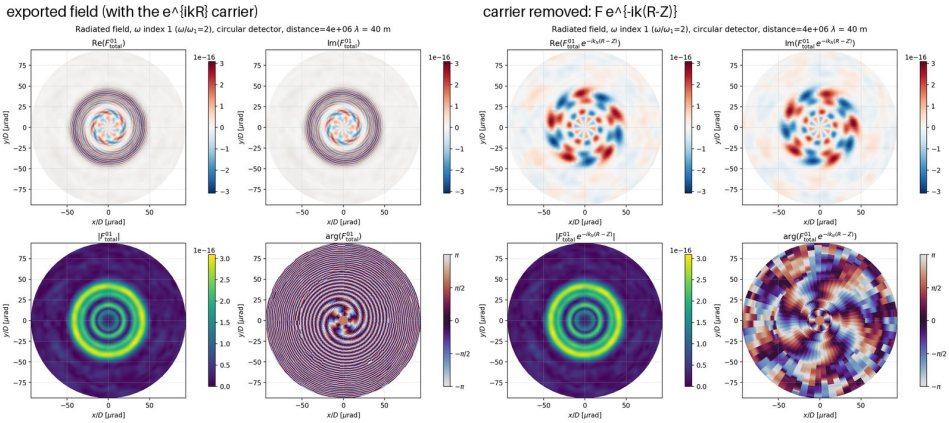


Figure 1: Run C (LG_2^2, $a_0 = 0.1$), harmonic $N = 2$, component F^{01} ($\propto E_x$). Left: the field as computed. Right: the same field with the spherical carrier removed. Each panel shows Re, Im, modulus and phase over the screen (axes: angle from the axis in μrad).

Things to notice in Fig. 2:

- **Gaussian laser, $N = 1$:** a single bright spot with no phase winding. The emitted beam is Gaussian.

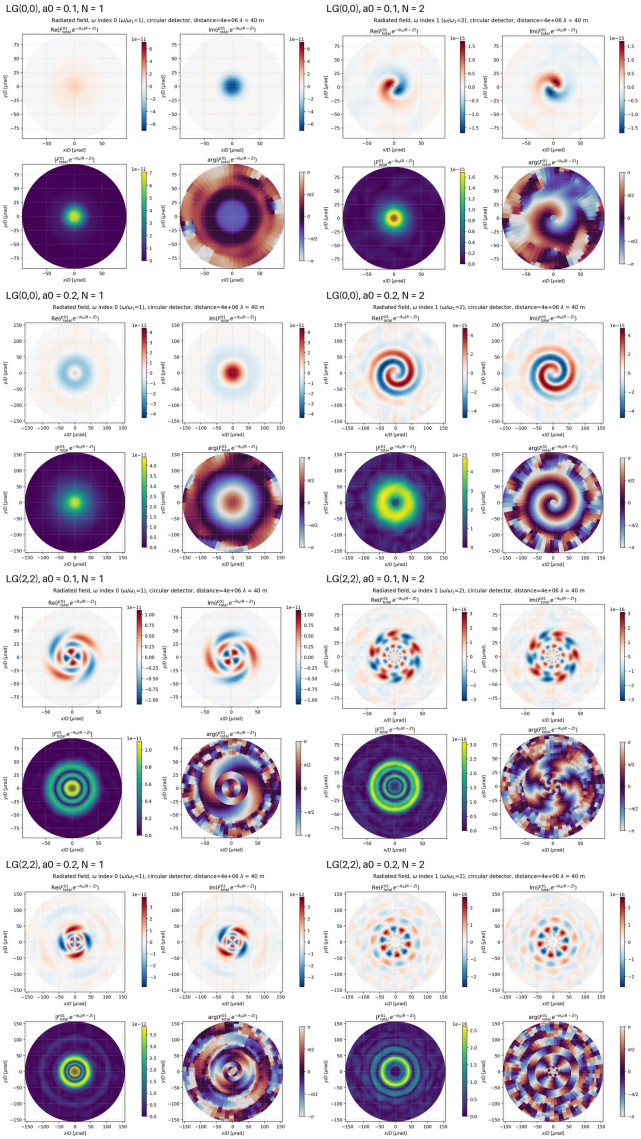


Figure 2: F^{01} with the carrier removed, for all four runs (rows A, B, C, D) and both harmonics (columns $N = 1, 2$).

- **Gaussian laser,** $N = 2$**:** a ring with one phase winding around the axis: the second harmonic carries orbital angular momentum, although the laser carries none. It comes from the laser's spin (Section 3).
- LG_2^2 **laser,** $N = 1$**:** three bright rings, the radial structure of the $p = 2$ laser mode, with a two-fold phase winding.
- LG_2^2 **laser,** $N = 2$**:** a phase winding of order five, and a radial structure that is no longer that of a single mode (Section 4).
- **From** $a_0 = 0.1$ **to** 0.2, the Gaussian-laser beams become wider on the screen (note the larger axes). The stronger field acts on the emission like a lens.

3. Observables: energy and angular momentum

3.1 Time-integrated spectral densities

The field is known at one frequency, so we use time-integrated quantities per unit frequency. For two real fields $A(t)$, $B(t)$ with Fourier amplitudes $\tilde{A}(\omega)$, $\tilde{B}(\omega)$, the time integral of the product $A_i B_j$ has the one-sided spectral density

$$\frac{d\langle A_i B_j\rangle}{d\omega} = 2\,\mathrm{Re}[\tilde{A}_i(\omega)\,\tilde{B}_j^*(\omega)], \qquad \omega > 0.$$

We use the radiation gauge, in which $\tilde{\mathbf{A}} = \tilde{\mathbf{E}}/(i\omega)$. All observables then follow from $\mathbf{E}$ and $\mathbf{B}$ on the screen.

3.2 Definitions

Energy. The energy density and the energy flux through the screen (Poynting vector along z) are

$$\frac{du}{d\omega} = \epsilon_0(|\tilde{\mathbf{E}}|^2 + c^2|\tilde{\mathbf{B}}|^2), \qquad \frac{dP_z}{d\omega} = 2\epsilon_0 c^2\,\mathrm{Re}[\tilde{E}_x\tilde{B}_y^* - \tilde{E}_y\tilde{B}_x^*].$$

Spin angular momentum. The spin density is $\mathbf{S} = \epsilon_0\,\mathbf{E}\times\mathbf{A}$. Its z component is

$$\frac{dS_z}{d\omega} = \frac{4\epsilon_0}{\omega}\,\mathrm{Im}[\tilde{E}_x^*\tilde{E}_y].$$

Orbital angular momentum. We use the canonical OAM density, $L_i = \epsilon_0 \sum_m E_m(\mathbf{r}\times\nabla)_i A_m$. Its z component contains the operator $(\mathbf{r}\times\nabla)_z = x\partial_y - y\partial_x = \partial_\phi$, which measures how fast the phase winds around the axis:

$$\frac{dL_z}{d\omega} = \frac{2\epsilon_0}{\omega} \sum_i \mathrm{Im}\Big[\tilde{E}_i^* \frac{\partial \tilde{E}_i}{\partial \phi}\Big].$$

For a field $\propto e^{il\phi}$, this is l/ω times the energy density, i.e. $l\hbar$ per photon. The spin and orbital parts together give the total angular momentum $J_z = S_z + L_z$.

Fluxes through the screen. On a screen, what is measured is the flux, not the density. Spin and orbital angular momentum each obey their own continuity equation in vacuum, $\partial_t S_z + \partial_j \Sigma_{zj} = 0$ and $\partial_t L_z + \partial_j \Lambda_{zj} = 0$. The fluxes along the screen normal are

$$\frac{d\Sigma_{zz}}{d\omega} = \frac{2\epsilon_0 c^2}{\omega} \mathrm{Im}[\tilde{B}_x^* \tilde{E}_x + \tilde{B}_y^* \tilde{E}_y - \tilde{B}_z^* \tilde{E}_z],$$

$$\frac{d\Lambda_{zz}}{d\omega} = \frac{2\epsilon_0 c^2}{\omega} \mathrm{Im}\Big[\tilde{B}_y^* \frac{\partial \tilde{E}_x}{\partial \phi} - \tilde{B}_x^* \frac{\partial \tilde{E}_y}{\partial \phi} + \tilde{B}_z^* \tilde{E}_z\Big].$$

We integrate every quantity over the screen. As a check, each flux divided by the corresponding density comes out as c in all runs ($\Sigma_{zz}/S_z = \Lambda_{zz}/L_z = P_z/u = c$ to six digits), as expected for radiation moving outward at the speed of light.

Angular momentum per photon. A photon of frequency ω carries energy $\hbar\omega$. So $\omega\Sigma_{zz}/P_z$ and $\omega\Lambda_{zz}/P_z$ are the spin and the orbital angular momentum per photon, in units of $\hbar$.

3.3 What we expect

Angular momentum about the collision axis is conserved. In harmonic N, one emitted photon takes the angular momentum of N absorbed laser photons. About the lab $+z$ axis (the electrons' direction), each laser photon carries orbital $m = -l$ and spin $\sigma = -1$, because the laser propagates along $-z$. Hence

$$J_z = N\,(m + \sigma) \quad \text{per emitted photon.}$$

If the emitted photon has spin -1, its orbital part is $J_z + 1$. So the Gaussian laser ($m = 0$) should give orbital 0 at $N = 1$ and -1 at $N = 2$. The LG_2^2 laser ($m = -2$) should give -2 and -5.

3.4 Results

Angular momentum per photon, integrated over the screen (units of $\hbar$):

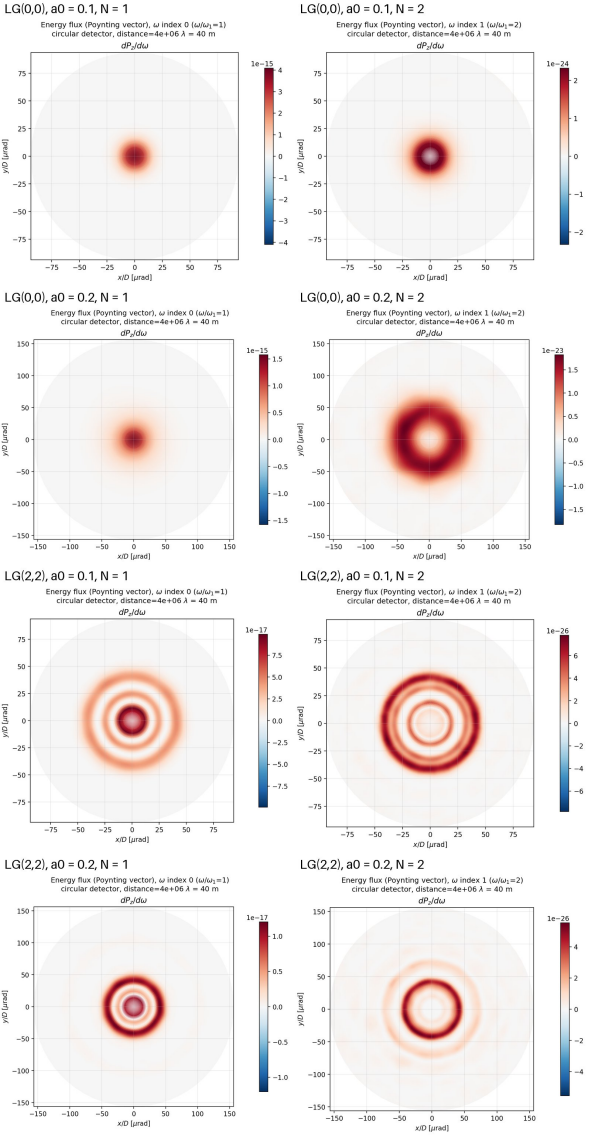


Figure 3: Energy flux through the screen, $dP_z/d\omega$, for all runs (rows A, B, C, D) and both harmonics (columns $N = 1, 2$).

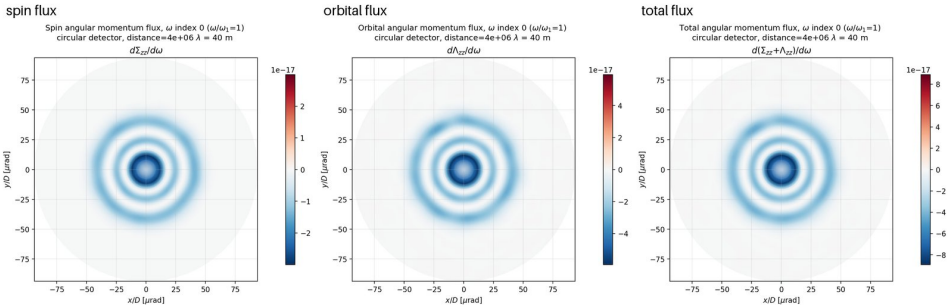


Figure 4: Run C (LG_2^2, $a_0 = 0.1$), $N = 1$: spin flux, orbital flux and their sum. The two have the same shape, and the orbital flux is twice the spin flux, as expected for photons with spin -1 and orbital -2.

Run	N	spin	orbital	total	expected total $N(m+\sigma)$
A: LG_0^0, $a_0 = 0.1$	1	-1.000	-0.000	-1.000	-1
	2	-1.000	-1.014	-2.014	-2
B: LG_0^0, $a_0 = 0.2$	1	-1.000	-0.001	-1.001	-1
	2	-1.000	-1.022	-2.022	-2
C: LG_2^2, $a_0 = 0.1$	1	-1.000	-1.991	-2.991	-3
	2	-1.000	-4.951	-5.951	-6
D: LG_2^2, $a_0 = 0.2$	1	-1.000	-1.963	-2.963	-3
	2	-1.000	-5.196	-6.196	-6

- Every emitted photon has spin exactly -1: all harmonics are circularly polarized with a single helicity. The other helicity carries $\sim 10^{-12}$ of the power.
- The orbital angular momentum agrees with $N(m+\sigma)-\sigma$ to 1–4%. The LG_2^2, $a_0 = 0.2$ harmonics (run D) deviate most. On the lattice (Sec. 5) every value agrees to 10^{-4}: the deviations are sampling noise.
- The second harmonic of a Gaussian laser carries one unit of orbital angular momentum, converted from the laser's spin.

4. Decomposition into Laguerre–Gauss modes

4.1 What is fitted

The goal is to describe each emitted harmonic as a superposition of LG modes and to find their waist w_0 and waist position z_0. We work with the two helicity components of the transverse field,

$$E_\pm = \frac{E_x \mp i E_y}{\sqrt{2}},$$

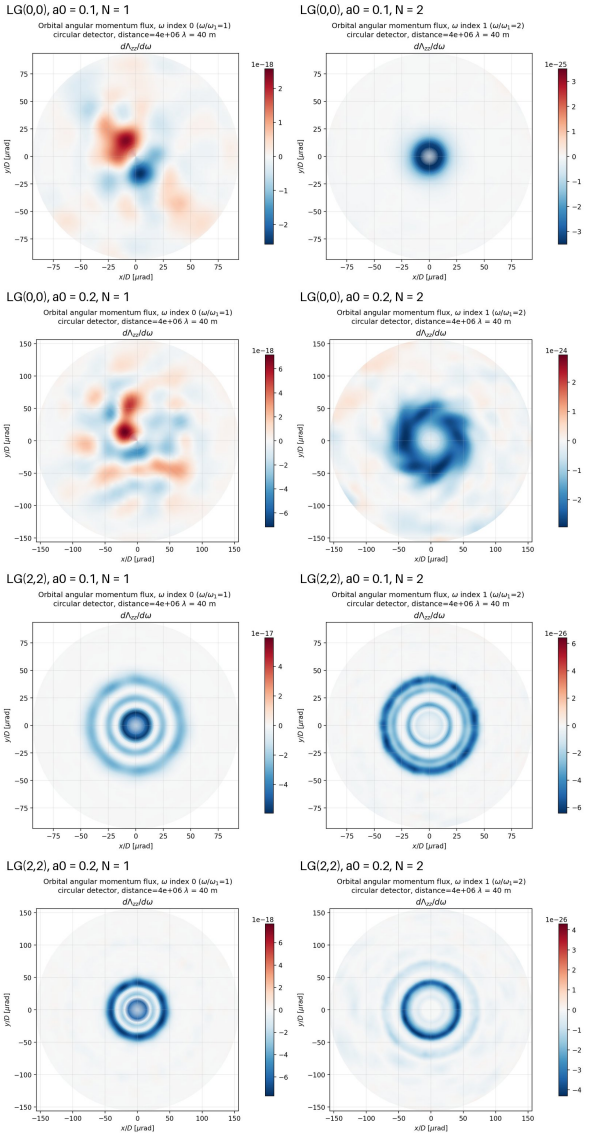


Figure 5: Orbital angular momentum flux $d\Lambda_{zz}/d\omega$ for all runs (rows A, B, C, D; columns $N = 1, 2$). For the Gaussian laser at $N = 1$ (left column, first two rows) the flux is only small positive and negative patches, which integrate to ≈ 0: this beam carries no orbital angular momentum.

which carry spin ± 1 about $+z$. All the power is in E_- (Section 3), so only E_- is shown below.

A component of fixed spin and fixed total angular momentum has one azimuthal index, $l = J_z - s = N(m + \sigma) + 1$. That gives $l = 0, -1$ (Gaussian laser) and $l = -2, -5$ (LG_2^2 laser) for $N = 1, 2$. We fit only the radial index p, using the normalized modes

$$u_{p,l}(r, \phi; w_0, z_0) \propto \frac{1}{w}\Big(\frac{\sqrt{2}\, r}{w}\Big)^{|l|} L_p^{|l|}\Big(\frac{2r^2}{w^2}\Big)\, e^{-r^2/w^2} \times e^{il\phi}\, e^{ikr^2/2R_c}\, e^{-i(2p+|l|+1)\psi},$$

evaluated on the screen at distance $s = Z - z_0$ from the waist, with $w(s)$, $R_c(s)$, $\psi(s)$ the usual beam width, wavefront radius and Gouy phase. Here $p = 0 \ldots 3$. The modes keep the full complex phase, so the fit uses the wavefront curvature as well as the intensity. The waist w_0 and the waist position z_0 are scanned on a grid (here 1.5–75 λ_L and $-1.5 \times 10^6 \ldots + 1.5 \times 10^6\, \lambda_L$). The z_0 range is kept well away from the screen, where the trial modes would become narrower than the grid.

4.2 Two methods

At every trial (w_0, z_0), the field E (here E_-) is described in the basis $\{u_{p,l}\}$ in two different ways. A_i are the area weights of the screen points, and $\|E\|^2 = \sum_i A_i |E_i|^2$ is the field's power on the screen.

Method 1: projections (overlap integrals).

$$a_p = \sum_i A_i\, u_{p,l}^*(\mathbf{x}_i)\, E(\mathbf{x}_i).$$

The modes are orthonormal over the infinite plane. If the field is contained on the screen, a_p is therefore exactly the expansion coefficient of the field, however much of a trial mode the finite screen cuts off. So $|a_p|^2/\|E\|^2$ is the share of the field's power in mode p. Each coefficient is a separate integral and does not depend on which other modes are included. We report two quantities from it:

- the **mode overlap**, $\max_p |a_p|^2/\|E\|^2$: the share of the field in the best single mode. It equals 1 only for a pure LG mode at its own w_0 and z_0;
- $\sum_p |a_p|^2/\|E\|^2$: the share of the field in all the fitted modes together.

Method 2: weighted least squares.

$$\min_c \sum_i A_i \Big| E(\mathbf{x}_i) - \sum_p c_p\, u_{p,l}(\mathbf{x}_i) \Big|^2.$$

This gives the best reconstruction of the field *on the screen* with these modes, and the relative residual ε measures how good it is. The two methods agree when the modes are orthonormal on the screen itself. When the trial waist is small, the modes are wide on the screen ($w(s) \propto 1/w_0$ in the far field) and get clipped. They are then no longer orthogonal on the screen, and the c_p lose their meaning as power fractions while the a_p keep it.

The two methods answer different questions: the projections say *how much of the beam is in each mode*; least squares says *how well the modes reproduce what the screen sees*. We take the (w_0, z_0) with the largest mode overlap (Method 1) as the beam's waist and waist position, and use the least-squares residual as a cross-check.

Independent check of l. On each ring of the circular screen, a Fourier transform in ϕ gives the power at every azimuthal index, without any assumption about the radial shape. Its share at the expected l is the "l purity" in the table below.

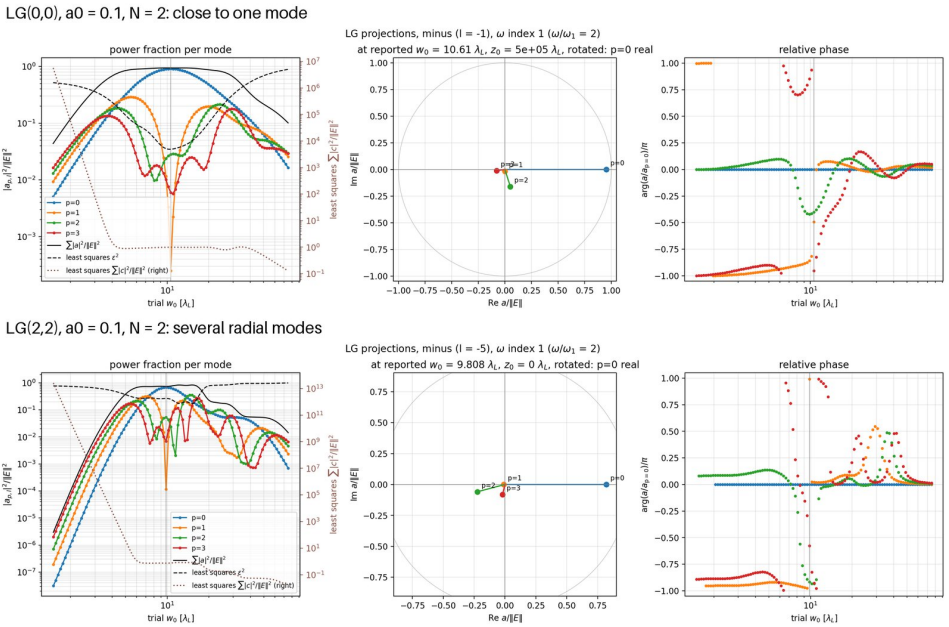


Figure 6: Both methods as a function of the trial waist, at the best z_0. Left: power fraction $|a_p|^2/\|E\|^2$ per mode (coloured), their sum (solid black), the least-squares ε^2 (dashed) and the least-squares $\sum |c|^2/\|E\|^2$ (dotted, right axis; far above 1 at small w_0, where the clipped modes are not orthogonal on the screen). Middle: the coefficients $a_p/\|E\|$ in the complex plane at the reported waist (unit circle = the whole field in one mode). Right: phases relative to $p = 0$. Top: Gaussian laser, $N = 2$: 91% of the power in the single mode LG_0^{-1}. Bottom: LG_2^2 laser, $N = 2$: the power is spread over several radial modes.

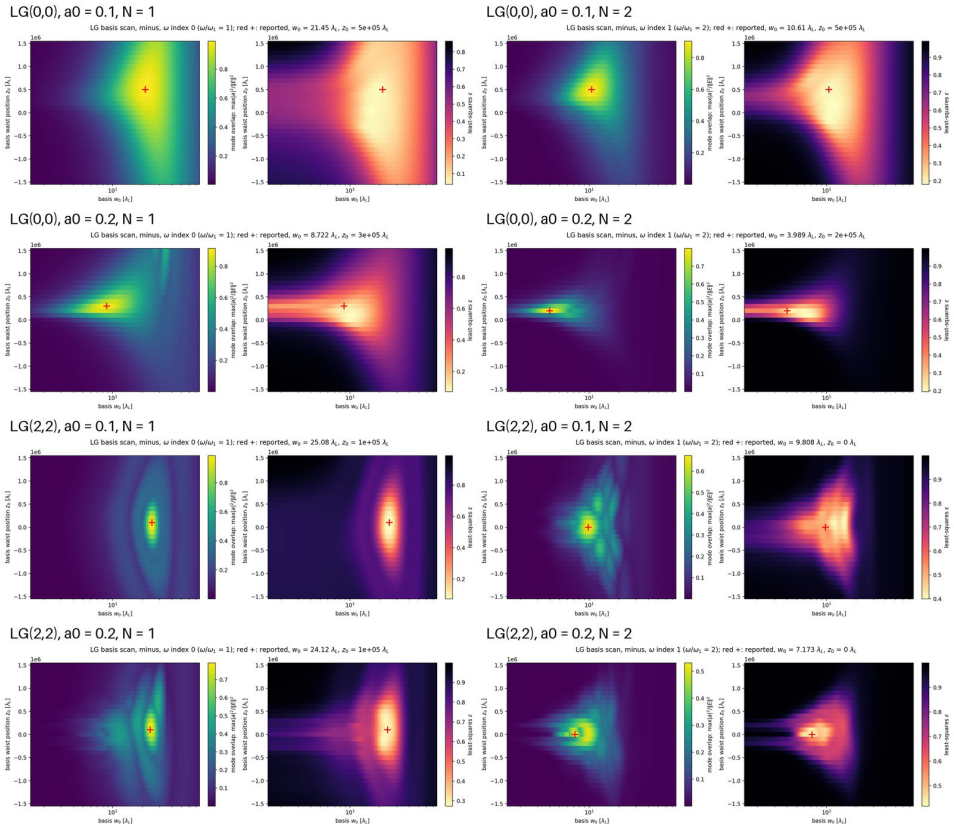


Figure 7: Mode overlap (Method 1, left of each pair) and least-squares residual ε (Method 2, right) over the trial (w_0, z_0) grid, for all runs (rows A, B, C, D) and harmonics (columns $N = 1, 2$). The cross marks the reported point. Where the field is close to one mode, both methods find a single, well-defined optimum at the same place.

4.3 Results (E_{-})

Run	N	l	l purity	best mode	share	w_0 [λ_L]	z_0 [λ_L]	all modes
A	1	0	0.999	LG_0^0	0.994	21.4	5×10^5	1.01
	2	−1	0.972	LG_0^{-1}	0.905	10.6	5×10^5	0.94
B	1	0	0.997	LG_0^0	0.915	8.7	3×10^5	0.99
	2	−1	0.968	LG_0^{-1}	0.800	4.0	2×10^5	0.94
C	1	−2	0.996	LG_2^{-2}	0.987	25.1	1×10^5	0.99
	2	−5	0.971	LG_0^{-5}	0.674	9.8	0	0.73
D	1*	−2	0.987	LG_2^{-2}	0.793	24.1	1×10^5	0.89
	2	−5	0.911	LG_1^{-5}	0.531	7.2	0	0.83

Runs: A = LG_0^0, $a_0 = 0.1$; B = LG_0^0, $a_0 = 0.2$; C = LG_2^2, $a_0 = 0.1$; D = LG_2^2, $a_0 = 0.2$. "Share" is the mode overlap, the share of the power in the best mode; "all modes" is $\sum_p |a_p|^2/\|E\|^2$, the share in $p = 0\,...\,3$ together. w_0 is in units of the laser wavelength λ_L. The emitted wavelength is about 700 times shorter, so the same waist is about 700 emitted wavelengths. z_0 is measured from the interaction point toward the screen ($Z = 4\times10^6\,\lambda_L$). *This fit is flagged as under-resolved: the beam covers too few radial grid points of the large screen, so its share is uncertain.

What the decomposition shows:

- **The azimuthal index is the one angular-momentum conservation predicts** in every case, with 91–100% of the power at that l.
- **Gaussian laser:** both harmonics are close to a single LG mode, LG_0^0 at $N = 1$ and LG_0^{-1} at $N = 2$ (91–99% at $a_0 = 0.1$). At $a_0 = 0.2$ the emitted waist is much smaller ($21 \to 9\,\lambda_L$ at $N = 1$) and the waist moves closer to the interaction point. This is the lens-like effect of the ponderomotive dressing that is visible in Fig. 2 (explained in Sec. 4.4). At the larger a_0 the share in one mode also drops, to 80–92%.
- LG_2^2 **laser,** $N = 1$**:** the emitted beam copies the laser mode, LG_2^{-2} with $w_0 \approx 25\,\lambda_L$, the laser's own waist (99% at $a_0 = 0.1$).
- LG_2^2 **laser,** $N = 2$**:** clean $l = -5$, but not a single LG mode. The power is spread over several radial indices (Fig. 6, bottom), and at most 67% is in any one mode. This is expected: the N-th harmonic is emitted by a source proportional to the N-th power of the laser mode, and the square of LG_2^2 is not itself an LG mode.

4.4 The ponderomotive lens

The emitted beams of the Gaussian laser look as if they had passed through a focusing lens. Their apparent waist is smaller than the laser's and sits downstream of the interaction point, and more so at $a_0 = 0.2$ (Sec. 4.3). The cause is the ponderomotive drift.

Physical picture.

- **The disk dents.** Inside the pulse, every electron drifts along the laser direction by an amount proportional to the local intensity $a^2(r)$. The flat electron disk becomes a dent shaped like the laser's intensity profile: deepest on the axis, and deepening steadily while the pulse passes. The dent is shallow, about a tenth of a wavelength.
- **Each electron's emission is locked to the laser phase.** It emits N harmonic cycles per laser cycle, wherever it is.
- **The drift delays the emission.** Seen from the screen, which faces the laser, a drifting electron's cycles arrive stretched, which is the familiar ponderomotive redshift, and delayed. At a fixed frequency the delay is a phase.
- **The axis lags most, so the disk acts as a focusing lens,** like glass that is thicker in the middle. The dent follows the Gaussian intensity profile rather than a parabola, so the lens is aberrated. That is why the emitted beam is no longer a single LG mode.

Model. For an electron at radius r, the drift multiplies its emission at the frequency ω we evaluate by

$$d_N(r) = \frac{\int d\xi\, g(\xi)^N\, e^{\,i\Phi_N(r,\xi)}}{\int d\xi\, g(\xi)^N}, \qquad \Phi_N(r,\xi) = \int_{\xi_0}^{\xi} \Big[\frac{\omega}{\omega_L}(1 + A(r,\xi')) - N\Big]\, d\xi',$$

where:

- ξ is the laser phase at the electron, and g the pulse envelope;
- $A = \langle a^2 \rangle$ is the local intensity at the electron;
- $1 + A$ is the stretching of the emitted wave train.

The phase of d_N is the delay. Its magnitude drops below 1 where the electron's redshifted frequency differs from the ω we evaluate at. That ω is resonant on a ring at $r \approx 0.6\, w_0$, not on the axis, because the LG mode's on-axis field is $\sqrt{2}$ times the plane-wave value used for ω (Sec. 7).

For electrons at rest the emitted field can then be predicted without any free parameter. The source of harmonic N is the laser mode to the N-th power, times $d_N(r)$, over the disk. We propagate it to the screen (Fresnel diffraction) and multiply by each electron's own angular factor. We compare it with the simulation through the **fidelity**, the share of the simulated field's power that the prediction explains once an overall complex factor is fitted.

Test. The setup:

- the laser of runs A and B: LG_0^0 with $w_0 = 25\,\lambda_L$ and the same pure Gaussian pulse, at $a_0 = 0.01$, 0.1 and 0.2;
- electrons at rest on a flat disk of the same radius as in runs A and B, sampled on the lattice of Sec. 1;

- a screen facing the laser at $Z = 2 \times 10^4\,\lambda_L$, about 10 Rayleigh ranges.

Electrons at rest avoid the Doppler mapping of a moving beam, so the model applies directly. Fidelity of the prediction for E_-:

	$N = 1$	$N = 2$
$a_0 = 0.01$, without / with d_N	0.9999 / 0.9999	0.998 / 0.998
$a_0 = 0.1$, without d_N	0.922	0.564
$a_0 = 0.1$, with d_N	**0.9999**	**0.9986**
$a_0 = 0.1$, phase of d_N only	0.9986	0.9966
$a_0 = 0.2$, without d_N	0.332	0.016
$a_0 = 0.2$, with d_N	**0.9996**	**0.997**
$a_0 = 0.2$, phase of d_N only	0.866	0.835

What the test shows:

- **With the dressing factor, the prediction reproduces the simulated field** to 99.7–99.99%, at both field strengths and both harmonics, with no free parameter. Without it, it explains 92% ($N = 1$) and 56% ($N = 2$) at $a_0 = 0.1$, and only 33% and 2% at $a_0 = 0.2$. At $a_0 = 0.01$ the factor changes nothing, as it should.
- **The lens is the delay phase.** At $a_0 = 0.1$ the phase of d_N alone gives almost everything. At $a_0 = 0.2$ the magnitude matters too (phase alone: 87% and 84%). Electrons then emit far enough from the evaluated frequency that their contribution is visibly weakened.
- **On the screen** (Fig. 8, $a_0 = 0.1$):
  - $N = 1$**:** the intensity barely changes. The wavefront, though, is that of a beam whose apparent focus sits about $700\,\lambda_L$ downstream: 1.3 rad more phase at 25 mrad.
  - $N = 2$**:** the phase is twice as large (N times). The ring keeps its radius (7 mrad) but gains a tail out to 25 mrad.
- **The effect does not fade with distance.** The far field is the Fourier transform of the source, so a phase of a few radians across the source changes the angular pattern, in amplitude and phase, at any distance.
- **It grows as $N\,a_0^2$ times the pulse length:** the dent deepens for as long as the pulse passes. Doubling a_0 makes the phase four times larger, which is why the prediction without d_N fails almost completely at $a_0 = 0.2$.

Runs A–D have moving electrons, which are slowed down rather than pushed forward, but the mechanism is the same. Extending the model to a moving beam, through the Doppler mapping of the source, is the next step.

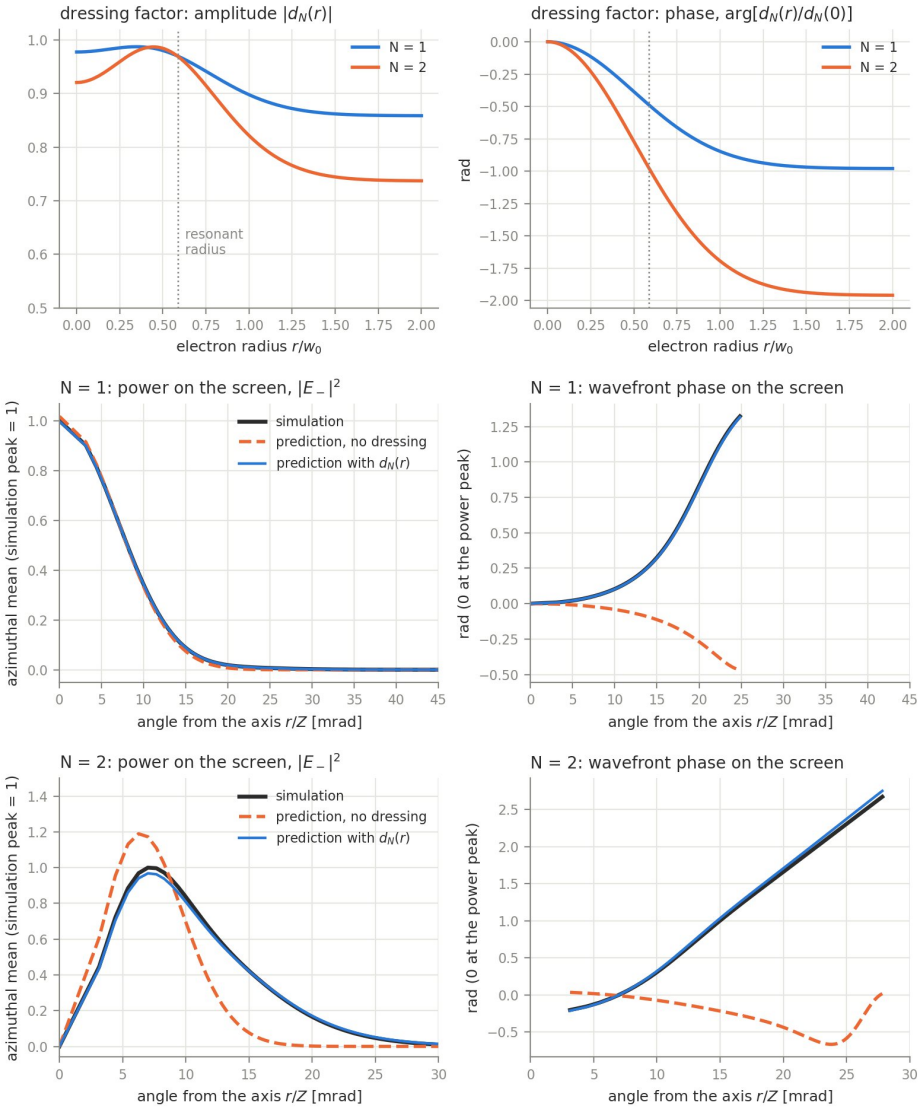


Figure 8: The ponderomotive lens: electrons at rest, the pulse of runs A–D, $a_0 = 0.1$. Top: the dressing factor of one electron, magnitude (left) and phase relative to the axis (right); the dotted line marks the resonant radius. Middle and bottom: the power $|E_-|^2$ on the screen (left; azimuthal mean) and its wavefront phase (right; along one ray, after removing the curvature of a beam focused at the interaction point) for $N = 1$ and $N = 2$. Black: simulation. Orange, dashed: prediction without the dressing factor. Blue: with it. Each prediction has one fitted complex factor.

5. Lattice versus random sampling

Runs A–D place 65 536 electrons at random on the disk, and their fields carry the speckle described in Sec. 1. We repeated all four with the electrons on the lattice of Sec. 1 (spacing $d \approx 2.5\,\lambda_L$). Everything else is unchanged: laser, beam, screen and post-processing. The lattice needs 1 076 electrons for the LG_0^0 disk and 3 319 for the larger LG_2^2 disk, 60 and 20 times fewer, and a run takes 40–110 s on 256 cores instead of 37 min. As a convergence check, run D was repeated with d halved (13 070 electrons). The field changes by 0.4% ($N = 1$) and 1.8% ($N = 2$) in amplitude, that is, 0.002% and 0.03% of the power.

5.1 The speckle of the random runs

The two fields of each run are compared point by point on the screen: the lattice field F_{lat} is scaled by the best complex factor c (close to the ratio of the electron numbers), and η is the share of the random field's power that it does not explain:

$$\eta = \frac{\|F_{\mathrm{rand}} - c\,F_{\mathrm{lat}}\|^2}{\|c\,F_{\mathrm{lat}}\|^2},$$

summed over the screen and all six components of F.

Run	η at $N = 1$	η at $N = 2$
A: LG_0^0, $a_0 = 0.1$	0.16%	4.2%
B: LG_0^0, $a_0 = 0.2$	0.39%	3.4%
C: LG_2^2, $a_0 = 0.1$	0.43%	3.2%
D: LG_2^2, $a_0 = 0.2$	1.4%	10%

η is the speckle, the artificial incoherent part of the random runs. It is larger at $N = 2$ and on the larger screen of run D, which resolves more modes. For run B, the bunch-length test (Sec. 6) found the same values (0.3% and 3%) on a different random sample.

5.2 Angular momentum

Orbital angular momentum per photon (units of $\hbar$; spin is -1.000 in every case):

Run	N	random	lattice	expected $N(m+\sigma)-\sigma$
A	1	-0.0001	0.0000	0
	2	-1.0139	-1.0000	-1
B	1	-0.0009	0.0000	0
	2	-1.0215	-0.9999	-1
C	1	-1.9914	-2.0000	-2

Run	N	random	lattice	expected $N(m+\sigma)-\sigma$
	2	-4.9511	-5.0000	-5
D	1	-1.9627	-2.0000	-2
	2	-5.1955	-5.0000	-5

On the lattice, angular-momentum conservation holds to 1.3×10^{-4} in every run. The 1–4% deviations of Sec. 3.4 were sampling noise, in run D as well.

5.3 LG decomposition

Same analysis as Sec. 4.3 (E_-), random → lattice:

Run	N	l purity	best mode	share	w_0 [λ_L]	z_0 [$10^5\,\lambda_L$]	all modes
A	1	0.999 → 1.000	LG_0^0	0.994 → 0.995	21.4	5	1.01 → 1.01
	2	0.972 → 1.000	LG_0^{-1}	0.905 → 0.945	10.6	5	0.94 → 0.97
B	1	0.997 → 1.000	LG_0^0	0.915 → 0.923	8.7	3	0.99 → 1.00
	2	0.968 → 1.000	LG_0^{-1}	0.800 → 0.830	4.0	2	0.94 → 0.97
C	1	0.996 → 1.000	LG_2^{-2}	0.987 → 0.991	25.1	1	0.99 → 1.00
	2	0.971 → 1.000	LG_0^{-5}	0.674 → 0.692	9.8	0	0.73 → 0.76
D	1*	0.987 → 1.000	LG_2^{-2}	0.793 → 0.800	24.1	1	0.89 → 0.89
	2	0.911 → 1.000	LG_1^{-5}	0.531 → 0.581	7.2	0	0.83 → 0.91

Best mode, w_0 and z_0 are the same for both samplings. *Under-resolved, as in Sec. 4.3.

- **All the power is at the expected** l (purity above 0.9999). The 0.1–9% at other l in Sec. 4.3 was speckle.
- **The mode shares rise by the speckle fraction**, $\mathrm{share}_{\mathrm{lat}} \approx \mathrm{share}_{\mathrm{rand}}\,(1+\eta)$: e.g. run D at $N=2$, $0.531 \times 1.10 = 0.584$ against 0.581. The best mode, its waist and its waist position do not change.
- **The conclusions of Sec. 4.3 stand.** The Gaussian laser's harmonics are close to single LG modes (now 95–99.5% at $a_0 = 0.1$), the lens at $a_0 = 0.2$ is unchanged, and the LG_2^2 laser's $N=2$ is not a single mode (at most 69% in one mode). Run D's lower $N=1$ share (0.80) is not sampling either; it remains under-resolved on its screen.
- **The lattice is the better method for these runs**: exact angular momentum, no speckle, and 20–60 times cheaper.

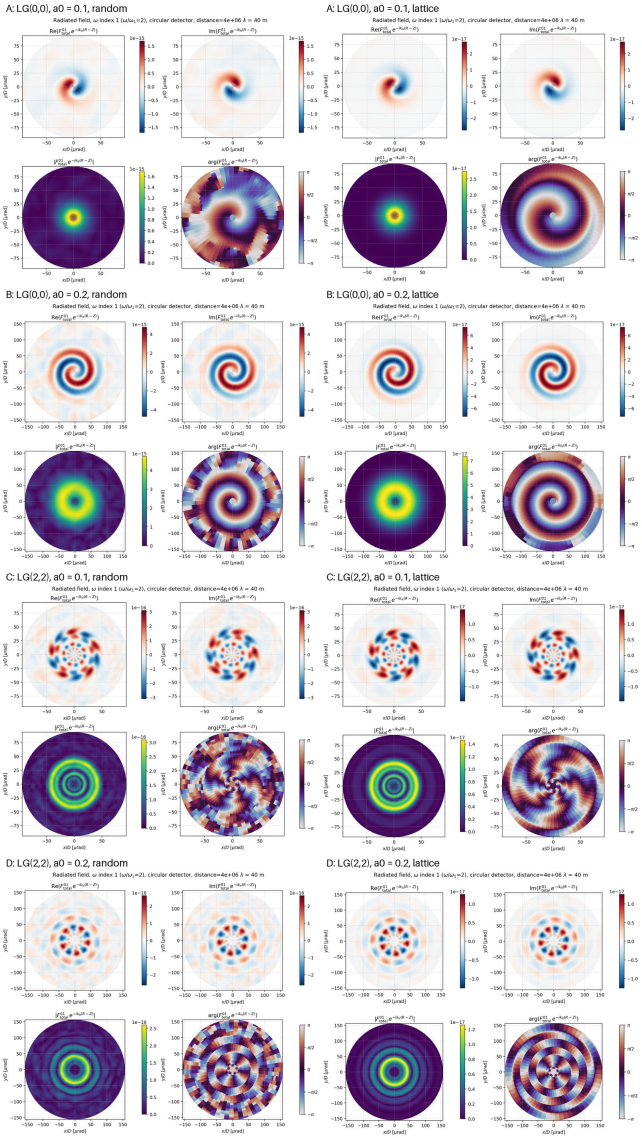


Figure 9: F^{01} at $N = 2$, carrier removed as in Fig. 2, for runs A–D (rows): random sampling (left) and the lattice (right). The lattice removes the speckle background and leaves clean phase spirals out to the edge of the screen.

Part II: a Gaussian bunch length

6. Loss of coherence in a bunch of finite length

6.1 What we expect

In Part I all electrons start in one plane. A real bunch also has a length. Take a Gaussian longitudinal distribution of r.m.s. length σ. As long as the bunch is much shorter than the laser's Rayleigh range ($z_R \approx 2000\,\lambda_L$), an electron that starts a distance z_0 behind the disk meets the same laser field a little later, so it follows the same trajectory shifted in time. Its emitted field is then the disk electron's field times a phase:

$$e(z_0) = e(0)\, e^{iq_N z_0}, \qquad q_N = Nk_1 + k_L,$$

where $k_1 = \omega_1/c$ and k_L is the laser wave number (head-on, on axis; across the screen, which spans $\sim 10^{-4}$ rad, q_N is constant to 10^{-8}). Averaging over z_0 gives the bunch form factor:

$$F_\sigma = \tilde\rho(q_N)\, F_0 + R, \qquad \tilde\rho(q) = e^{-q^2\sigma^2/2}.$$

- The **coherent field** keeps its shape (mode, waist, waist position, phase winding) and its amplitude falls by $\tilde\rho$, so its power falls by $\tilde\rho^2$.
- The rest, R, is **incoherent**: its power is $N\langle|e|^2\rangle\,(1-\tilde\rho^2)$, proportional to the number of electrons N rather than N^2.

A test of this picture is whether the lens of Sec. 4.4 survives the bunch length unchanged.

6.2 Setup

We repeat run B (LG_0^0, $a_0 = 0.2$, $\gamma \approx 13.5$ head-on, screen radius $25\,w_0$), with the 65 536 electrons spread along z with a Gaussian of r.m.s. length σ. The lengths are chosen to give $q_1\sigma = 0.25$, 0.5 and 1, that is $\sigma = 0.56$, 1.1 and 2.2 nm; at $N = 2$, $q_2\sigma$ is twice as large. A reference run with $\sigma \approx 0$ uses the same transverse positions, so the four runs can be compared point by point on the screen, without the speckle of Sec. 1 getting in the way. (The reference and run B are two different random samples of the same disk. They agree to within the speckle: mode share 0.919 against 0.915 at $N = 1$, 0.808 against 0.800 at $N = 2$.)

For each run and harmonic we fit the field to the reference, $F_\sigma \approx c\,F_0$, over the whole screen and all six components of F. The model predicts $c = \tilde\rho(q_N)$. The residual, divided by $1-\tilde\rho^2$, is η, the incoherent power of the sample relative to its coherent power.

6.3 Results

$q_1\sigma$	N	$q_N\sigma$	$\tilde{\rho}$	$\lvert c\rvert/\tilde{\rho}$	$\arg c$ [rad]	η
0.25	1	0.25	0.969	0.9994	0.003	0.0029
	2	0.50	0.883	0.997	0.008	0.030
0.5	1	0.50	0.883	0.997	0.005	0.0031
	2	1.00	0.607	0.989	0.014	0.033
1	1	1.00	0.607	0.988	0.009	0.0036
	2	2.00	0.136	0.935	−0.040	0.034

- **The coherent field falls as the form factor predicts**: $|c| = \tilde{\rho}$ to 1.2% up to $q\sigma = 1$. The small remaining deviation depends only on $q\sigma$, the same at $N = 1$ and $N = 2$. It comes from the one random draw of the z positions, not from physics, and stays within the bound set by the incoherent part.
- **The rest is the incoherent part**: η is the same in all runs (0.3% at $N = 1$, 3% at $N = 2$), so the residual grows exactly as $1 - \tilde{\rho}^2$.

The lens survives. LG decomposition of E_- (Sec. 4), with the waist w_0 and waist position z_0 of the single LG beam that best matches the field:

$q_1\sigma$	N	best mode	share	l purity	w_0 [λ_L]	z_0 [$10^5\,\lambda_L$]	$\lvert a\rvert/\lvert a_{\mathrm{ref}}\rvert$
0	1	LG_0^0	0.919	0.998	8.55	2.81	1
0.25	1	LG_0^0	0.920	0.997	8.56	2.81	0.969
0.5	1	LG_0^0	0.921	0.997	8.56	2.81	0.882
1	1	LG_0^0	0.917	0.991	8.57	2.78	0.601
0	2	LG_0^{-1}	0.808	0.976	4.05	2.05	1
0.25	2	LG_0^{-1}	0.802	0.967	4.02	2.06	0.880
0.5	2	LG_0^{-1}	0.764	0.919	4.03	2.06	0.600
1	2	LG_0^{-1}	0.263	0.331	4.04	2.06	0.126

- **Same mode, same waist, same waist position.** Even at $q_2\sigma = 2$, where the coherent part holds only a third of the power on the screen, w_0 moves by less than 1% and z_0 by about 1%. The mode amplitude scales by $\tilde{\rho}$.
- **Share and l purity fall only through the added incoherent power.** Adding the incoherent part to the reference share predicts 0.765 ($q_2\sigma = 1$; measured 0.764) and about 0.29 ($q_2\sigma = 2$; measured 0.263).

6.4 What this means for a real bunch

- **A Gaussian bunch length only multiplies the coherent field by** $e^{-q_N^2\sigma^2/2}$ and adds an incoherent part. The ponderomotive lens, and everything else in the coherent field, is unchanged as long as the bunch is much shorter than z_R.

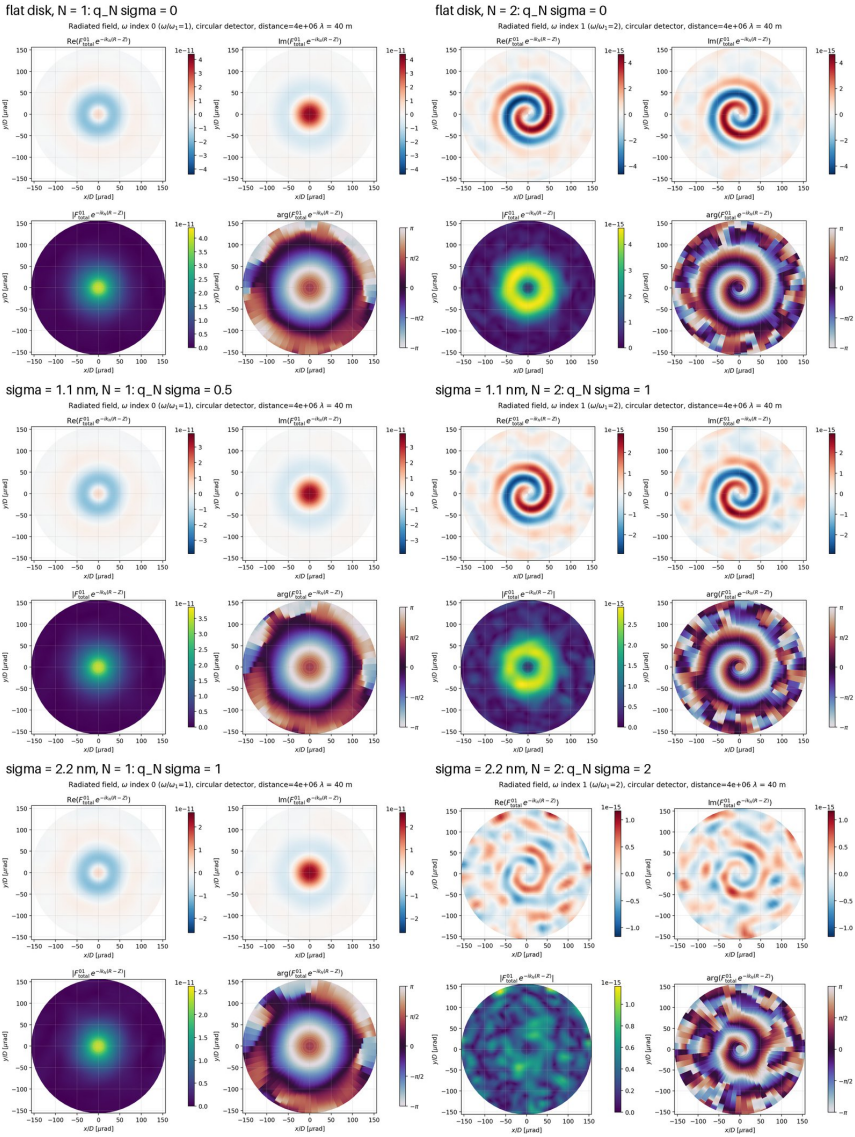


Figure 10: The field of Part II (F^{01}, carrier removed as in Fig. 2) for the flat disk and two bunch lengths (rows), at $N = 1$ and 2 (columns). At $N = 1$ the beam is unchanged, only weaker. At $N = 2$, $q_2\sigma = 2$ (bottom right), the coherent spiral is still on the axis at about 1/7 of its amplitude, and the incoherent speckle, of comparable size, covers the whole screen.

- **Coherence needs $\sigma \lesssim 1/q_N \approx 2$ nm** at these XUV harmonics ($\lambda_1 \approx 14$ nm). Real bunches are 3–30 $\lambda_L = 30$–300 μm long, $q_1\sigma \sim 10^4$–10^5: their coherent emission at these harmonics is zero for every practical purpose. That is why the third step, microbunching at the emitted wavelength, matters.
- **The incoherent part here belongs to the simulation, not to a real bunch.** The coherent power grows as N^2 and the incoherent as N, so η is that of 65 536 electrons. A bunch of 10^8–10^{10} electrons has an incoherent fraction 10^3–10^5 times smaller at the same $\tilde{\rho}$. The drops in share and purity in the table are therefore larger than a real bunch would show. To get both parts right, they must be computed separately, the coherent part from the form factor and the incoherent part from single-electron fields; that is planned.

7. Limitations of these results

- One frequency per harmonic, the on-axis one. Electrons at different radii are dressed differently and emit at slightly different frequencies (Sec. 4.4); at one frequency this shows as the lens, but how the mode content changes across the emission line is not followed yet.
- Sec. 2–4 use runs A–D, which sample the electrons at random and carry a speckle background (0.2–1.4% of the power at $N = 1$, 3–10% at $N = 2$). Sec. 5 repeats them on the lattice, without it; Part II is still random.
- An idealized electron beam: no energy or angular spread. Part I uses a zero-thickness disk, fully coherent; Part II adds only a Gaussian length, tested at nanometre lengths, far shorter than real bunches.
- In Part II the incoherent part is that of the 65 536 simulated electrons, too large by the ratio of real to simulated electrons (Sec. 6.4). The bunch-length test uses the undressed q_N; the dressed value differs by 6×10^{-5} relative, below what the test resolves.
- With our LG normalization, the Gaussian mode's on-axis field is $\sqrt{2}$ times that of a plane wave with the same a_0, so "a_0" is not quite the peak field strength.
- At $a_0 = 0.2$ the LG_2^2, $N = 1$ beam is under-resolved on the $25\,w_0$ screen. A rerun on a smaller screen would settle whether its lower mode share (0.79, against 0.99 at $a_0 = 0.1$) is physical. It is not sampling: the lattice gives the same 0.80 (Sec. 5).